PHYS 210 Electricity and Magnetism

Celebration of Knowledge!

Supplementary Practice 9 - CK #2 preparation

Name: _____ Date: _____

Time: _____ Marks: _____ Practice paper - not an official exam

Attempt all questions. Supporting practice; this topic mix is not the announced main-test format.

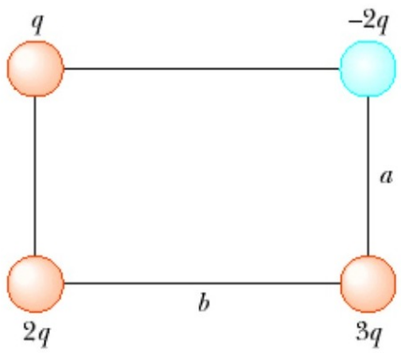
Write all rough calculations and final answers on separate paper. Show your working.

Original source numbering is retained inside each question. Answer the entire cropped question.

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1. [____] Tutorial 5 - Question 34

34. Calculate the energy required to assemble the array of charges shown in the figure below, where $a = 0.200$ m, $b = 0.400$ m, and $q = 6.00\ \mu$C.



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2. [____] Tutorial 5 - Question 39

39. Over a certain region of space, the electric potential is $V = 5x - 3x^2y + 2yz^2$. Find the expressions for the x, y, and z components of the electric field over this region. What is the magnitude of the field at the point P that has coordinates (1, 0, –2) m?

3. [____] Tutorial 5 - Question 42

42. Consider a ring of radius R with the total charge Q spread uniformly over its perimeter. What is the potential difference between the point at the center of the ring and a point on its axis a distance $2R$ from the center?

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4. [____] Tutorial 5 - Question 63

63. From Gauss's law, the electric field set up by a uniform line of charge is

$$E = \lambda / 2\pi\varepsilon_0 r$$

where λ is the linear charge density along the line. Derive an expression for the potential difference between $r = r_1$ and $r = r_2$.